

**Indian Institute of Technology Guwahati**  
**Department of Electronics and Electrical Engineering**  
**EE101 - Basic Electronics**

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**Solution: Pre-Tutorial Assignment - 8      Date: Oct 12, 2023      Marks: 2**

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1. The VI characteristic of a Germanium diode is given by

$$I_D = I_S(e^{\frac{V_D}{V_T}} - 1)$$

Derive and calculate the ac resistance  $r_d = \frac{dV_D}{dI_D}$  when  $V_T = 26 \text{ mV}$  and  $I_D = 1 \text{ mA}$

**Solution**

Taking the derivative of forward bias voltage with respect to the forward bias current, we get

$$\begin{aligned}\frac{d(I_D)}{dV_D} &= \frac{d}{dV_D}[I_S(e^{\frac{V_D}{V_T}} - 1)] \\ &= \frac{1}{V_T}(I_D + I_S)\end{aligned}$$

In general  $I_D \gg I_S$ , hence

$$\frac{dI_D}{dV_D} = \frac{I_D}{V_T}$$

Flipping the result to define a resistance ratio gives

$$r_d = \frac{dV_D}{dI_D} = \frac{V_T}{I_D}$$

Substituting the values of  $V_T$  and  $I_D$ , we get the ac resistance as

$$r_d = \frac{26 \text{ mV}}{1 \text{ mA}} = 26 \Omega$$